

PluralTree: Measurement, Validity and Identification

Companion to `methodology.tex`. That document defines the method; this one defines what its measurements identify, and states the conditions under which a reported number means what it appears to mean. Each section exists because the corresponding confusion actually occurred and cost a run.

1 Notation

Q is a set of questions. For $x \in Q$, $W(x)$ is its set of human viewpoint clusters and $F(x)$ the forks the scout retrieves. A condition c maps x to an answer $y_c(x)$, stochastically. $R_c(x) \subseteq W(x)$ is the set of clusters the judge marks covered by $y_c(x)$, and

$$\text{cov}_c(x) = |R_c(x)| / |W(x)|.$$

Conditions come in two families: the *control* C (typically BASELINE, no retrieval) and *treatments* T that consume $F(x)$.

2 Three measurement operators, and why they are not interchangeable

Three quantities in this project all read as “did the content get through” and measure different things. Conflating them produced two wrong conclusions.

Definition 1 (Coverage). $\text{cov}_c(x)$ above: a *judge* decides, per cluster, whether the holders of that viewpoint would say the answer represented them. Threshold on a predicted 1–5 rating.

Definition 2 (Usage). For injected positions $p \in P(x)$ and answer units $u \in U(y)$,

$$\text{usg}_\theta(y, x) = \frac{1}{|P(x)|} \sum_{p \in P(x)} \mathbf{1} \left[\max_{u \in U(y)} \cos(e(p), e(u)) \geq \theta \right].$$

A *surface* measure: did text resembling p appear. It is correlational — the model may have produced that unit regardless — and has no chance floor unless one is computed by permuting x against y from other questions.

Definition 3 (Retention). For drafts $\{y_i\}$ with deduplicated deep units D , and merged answer $\hat{y}$,

$$\text{ret}_\theta(\hat{y}, \{y_i\}) = \frac{1}{|D|} \sum_{d \in D} \mathbf{1} \left[\max_{u \in U(\hat{y})} \cos(e(d), e(u)) \geq \theta \right].$$

Measures whether *compression* destroyed content, not whether the content was ever correct or scored.

Remark 1 (The dissociation is empirical, not hypothetical). Measured: $\text{usg}_{0.35} \approx 0.61$ with no coverage gain; $\text{ret}_{0.55}$ median 1.00 while cov fell far below control. Surface presence, faithful preservation, and scored coverage move independently. A claim about one transfers to another only with an argument, never by default.

Remark 2 (Losslessness is not judge-invariant). Concatenating drafts is lossless in text: $\text{ret} = 1$ by construction. It is not lossless in coverage, because the judge scores a response *as a whole* — a good draft appended to two poor ones is rated as a mostly-poor response. So $\text{ret}(\hat{y}) = 1 \not\Rightarrow \text{cov}(\hat{y}) \geq \max_i \text{cov}(y_i)$.

3 Abstention and attenuation

When the scout resolves no anchor, $F(x) = \emptyset$ and the multi-draft condition keeps only its uninjected draft, so T reduces *exactly* to C in construction. Let $Q = Q_r \cup Q_a$ split Q into resolved and abstained, with abstention rate $\pi = |Q_a|/|Q|$.

Proposition 1 (Attenuation). *If $\mathbb{E}[\text{cov}_T(x) - \text{cov}_C(x)] = 0$ for all $x \in Q_a$, then*

$$\mathbb{E}[\hat{\delta}_{\text{pooled}}] = (1 - \pi) \delta_r, \quad \delta_r = \mathbb{E}_{x \in Q_r}[\text{cov}_T(x) - \text{cov}_C(x)].$$

Proof. $\hat{\delta}_{\text{pooled}} = |Q|^{-1} \sum_{x \in Q} (\text{cov}_T - \text{cov}_C)$. The Q_a terms have zero mean, leaving $|Q_r|/|Q| \cdot \delta_r$. $\square$

Remark 3 (Power loss is milder than attenuation suggests). Pooling attenuates the numerator by $(1 - \pi)$ but also adds $|Q_a|$ observations. If the per-question variance is comparable on both subsets ($\sigma_a \approx \sigma_r = \sigma$), then $\text{SE}_{\text{pooled}} \approx \sigma/\sqrt{|Q|}$ against $\text{SE}_r = \sigma/\sqrt{|Q_r|}$, so

$$\frac{t_{\text{pooled}}}{t_r} = (1 - \pi) \frac{\text{SE}_r}{\text{SE}_{\text{pooled}}} = (1 - \pi) \cdot \frac{1}{\sqrt{1 - \pi}} = \sqrt{1 - \pi}.$$

At $\pi = 0.29$ (INFINITY-CHAT): the effect estimate shrinks 29% but the t -statistic only 16%. Pooling is therefore *biased* rather than merely underpowered — the reported effect size is wrong, not just noisy.

Consequence. Any condition that can abstain must record whether it did, per row, and the resolved subset must be reported separately. Measured abstention rates: PERSONA_MERGE 21/180; MERGE_v2 on INFINITY-CHAT 29/100.

4 Ablation validity

An ablation compares T against T' differing in one component. If the component did not actually differ, the run produces a complete result table that means nothing: a tie is then *uninformative* rather than evidence of no effect. Validity conditions are therefore preconditions on the run, checked before its scores are read.

Let $n_c(x) = |F_c(x)|$ and $\bar{W}_c(x)$ the mean divergence of selected forks.

Definition 4 (Valid selection ablation). DIVRAND replaces the rank key $s(u, v)$ by a deterministic uniform draw over the *same* candidate pool. It is valid iff, over Q ,

$$(i) \mathbb{E}[n_{T'}] = \mathbb{E}[n_T], \quad (ii) \mathbb{E}[\bar{W}_{T'}] < \mathbb{E}[\bar{W}_T], \quad (iii) \mathbb{E}[\bar{\text{rel}}_{T'}] \approx \mathbb{E}[\bar{\text{rel}}_T].$$

(i) is structural: both modes rank one pool and truncate at k , so counts cannot diverge. (ii) is forced — T takes the argmax, so a uniform draw over the same pool must sit below it; equality means the draw selected the same pairs and the ablation is inert. (iii) distinguishes this from the *content* control, whose whole purpose is to be *less* relevant; applying that control’s check here would flag correct behaviour as failure.

Definition 5 (Valid hierarchy ablation). FLAT retrieves by similarity over opinion leaves with no descent. It is valid iff it abstains on the same questions: $F_{T'}(x) = \emptyset \iff F_T(x) = \emptyset$, enforced by applying the identical relevance gate τ . Without it the arms differ in *which questions get injected at all*, and Proposition 1 applies asymmetrically — confounding the hierarchy test with abstention.

Definition 6 (Valid content control). RAND substitutes structurally matched forks from an unrelated anchor. It is valid iff $\mathbb{E}[n_{T'}] = \mathbb{E}[n_T]$, rendered lengths match, and relevance is *lower*.

Remark 4 (This is not hypothetical either). A content control violated all three conditions for three consecutive runs (1.00 forks against 4.85, and a dispatch gap routing it to a different merge algorithm). Its reported tie was read as evidence about fork content; it was evidence about nothing.

5 Identification: which contrast measures which stage

The pipeline factors into RETRIEVE $\rightarrow$ DELIVER $\rightarrow$ COMPRESS. A change in cov can originate in any of them, and a null in one stage can be produced by a later stage destroying what an earlier stage supplied. Each contrast below holds two stages fixed.

contrast	identifies	holds fixed
$T_{c=0.5}$ vs. $T_{c=0}$	embedding geometry	graph, content, delivery, merge
T vs. DIVRAND	fork <i>selection</i>	graph, anchors, volume, merge
T vs. FLAT	<i>hierarchy</i>	volume, rendering, gate, merge
T vs. RAND	fork <i>content</i>	volume, structure, merge
α sweep (CAD)	delivery channel strength	prompt, sampler, RNG
7B vs. 72B	generator capacity	questions, drafts, retrieval
ret, guard on/off	compression fidelity	drafts
graph swap (ATP/ISSP)	corpus	eval, judge, questions, conditions
task swap (Overton/VITAL)	construct + scorer	graph, embeddings, conditions

Remark 5 (Graph swap and task swap are orthogonal). Changing the retrieval corpus while holding the eval fixed, and changing the eval while holding the corpus fixed, answer different questions. Neither is “more data” for the other: a replication of a conclusion on a different construct with a different scorer does not tighten the original confidence interval.

6 The noise floor from an invariant condition

BASELINE uses no retrieval. It is therefore *invariant* to every retrieval-side manipulation: embedding curvature, selection rule, hierarchy, fork content. Any difference in $\overline{\text{cov}}_C$ between two arms of such an ablation is pure run-to-run noise, and supplies a free scale against which the treatment delta is read.

Proposition 2 (Noise floor). *For arms g_1, g_2 differing only in a retrieval-side component, $\mathbb{E}[\overline{\text{cov}}_C^{(g_1)} - \overline{\text{cov}}_C^{(g_2)}] = 0$, so $|\overline{\text{cov}}_C^{(g_1)} - \overline{\text{cov}}_C^{(g_2)}|$ estimates the sampling scale of a single arm’s mean.*

Measured, each a single draw of the drift and therefore itself noisy: $|0.5089 - 0.4951| = 0.0138$ at 72B with one rollout per question; $|0.5024 - 0.5039| = 0.0015$ for the same two arms at three rollouts; and $|0.4277 - 0.3913| = 0.0364$ at 7B with one rollout. The floor depends on generator and rollout count, so it must be read from the run in question rather than quoted as one project-wide number. A treatment delta of the same magnitude as its own run’s baseline drift is not distinguishable from zero regardless of its nominal p -value.

7 Inference

Paired bootstrap over questions. For conditions c_1, c_2 on shared Q ,

$$\hat{\delta} = \frac{1}{|Q|} \sum_{x \in Q} [\text{cov}_{c_1}(x) - \text{cov}_{c_2}(x)],$$

with interval and two-sided p from 10^4 resamples *of questions*. Since resampling is over questions, between-question heterogeneity sets the floor on interval width: additional rollouts per question reduce only the within-question component, and cannot resolve an effect that question heterogeneity obscures. This is why $n = 60$, not rollout count, is the binding constraint.

Difference of differences. For arms g_1, g_2 , $\widehat{\Delta\Delta} = \hat{\delta}_{g_1} - \hat{\delta}_{g_2}$, bootstrapped paired over the same Q .

Remark 6 (The overlap fallacy). $\hat{\delta}_{g_1}$ significant and $\hat{\delta}_{g_2}$ not does *not* establish $\delta_{g_1} \neq \delta_{g_2}$. The geometry ablation at one rollout gave $+0.0136$ ($p = 0.62$) against $+0.0707$ ($p = 0.014$) — an apparently decisive contrast whose difference of differences was $p = 0.094$. At three rollouts the same arms gave $+0.0402$ and $+0.0453$, with $\widehat{\Delta\Delta} = +0.0051$, 95% CI $[-0.0402, +0.0573]$, $p = 0.88$: a well-powered null. The one-rollout picture was an artifact of within-question noise, in the direction that looked like a finding.

8 Union, oracle and gain

With $R_c(x)$ the covered set:

$$\begin{aligned} \text{union}(x) &= \left| \bigcup_{c \in S} R_c(x) \right| / |W(x)|, \\ \text{oracle}(x) &= \max_{c \in S} |R_c(x)| / |W(x)|, \\ \text{gain} &= \overline{\text{union}} - \max_{c \in S} \overline{\text{cov}_c}. \end{aligned}$$

$\text{union} \geq \text{oracle} \geq \max_c \overline{\text{cov}_c}$ holds by construction: adding an answer cannot remove a covered cluster, and a union is at least as large as the better single set. Note oracle is a mean of per-question maxima while $\max_c \overline{\text{cov}_c}$ is a maximum of means, so that inequality is arithmetic, not empirical.

Remark 7 (Gain measures difference, not quality). Gain is unrealised headroom, and is interpretable only jointly with the level: high level with low gain is the target; high level with high gain is good content under lossy compression; low level with high gain finds distinct viewpoints and delivers few. Critically, *low gain has two opposite causes* — an arm that already delivers everything it finds, and an arm that finds nothing new — which gain alone cannot separate. Arms equal in $\overline{\text{cov}}$ to within 0.006 have differed by $2.4\times$ in gain.

9 Scorer validity

A scorer used for *relative* comparison need not be unbiased, only *consistently* biased across the arms compared. This licenses far less than it appears to.

Definition 7 (Admissible comparison). A contrast c_1 vs. c_2 under scorer S is admissible if S 's bias is constant in every respect on which c_1 and c_2 differ.

Length is the binding case. If S is length-sensitive — documented for NLI-style coverage scorers, and unmeasured for the embedding-match reward — then any contrast whose arms differ systematically in answer length is inadmissible. Injected answers run roughly $5\times$ baseline's length, so BASELINE vs. a merge condition is inadmissible without a length control, while merge-vs-merge contrasts (geometry, selection, hierarchy) are admissible, having matched length distributions by construction.

Remark 8 (Validation level must match use level — and must be this judge's). Two separate gaps. First, a judge validated for *ranking systems* is not thereby validated for *per-question* agreement, and every within-question comparison in this project operates below the ranking level. Second, and more basic: the benchmark's reported $\rho = 0.88$ (rank correlation over 8 reference models) is the *benchmark authors'* judge. Reproduced with this project's judge, aggregate agreement is $\rho = 0.167$ (0.357 in an earlier run with too few participants per question) and within-participant $\rho = 0.059$. Whether that is poor depends on the human split-half ceiling at the same level, which has not been measured — “0.167 against 0.88” can look conclusive when the achievable ceiling sits far below 0.88. Until the ceiling is known the judge is unvalidated at every level, and its unmeasured reliability bounds any concordance a reward can reach against it.